

Soumya Tyagi

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Summary

AI/ML Engineer with hands-on experience in Machine Learning, Deep Learning, Natural Language Processing (NLP), and Generative AI. Proficient in Python, Scikit-learn, TensorFlow, PyTorch, FastAPI, Streamlit, LangChain, and Large Language Models (LLMs). Experienced in building end-to-end AI solutions involving data preprocessing, model training, evaluation, semantic matching, and AI-powered automation. Strong problem-solving skills with a solid foundation in Data Structures and Algorithms (DSA) and core computer science concepts.

Education

VIT Bhopal University

2023 – 2027

B.Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning), **CGPA: 8.43/10**

Technical Skills

Programming: Python, SQL, Java(DSA)

AI & Machine Learning: Machine Learning, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG)

Frameworks & Backend: LangChain, Scikit-learn, TensorFlow, PyTorch, FastAPI, Streamlit

Core CS: Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), DBMS

Experience

Videonza Technologies (OPC) Private Limited

May 2026 – Jul 2026

AI/ML Development Intern

Hybrid

- Preprocessed, analyzed, and transformed datasets to support machine learning model development and improve data quality.
- Trained and evaluated machine learning models using **Scikit-learn, TensorFlow, and PyTorch**, improving model accuracy from **82% to 91%** through data preprocessing and model optimization.
- Performed data cleaning, feature engineering, feature transformation, and exploratory data analysis (EDA) using Python and Scikit-learn to improve data quality and identify meaningful patterns.
- Supported **model evaluation, validation, and deployment workflows**, to improve the reliability and performance of AI/ML solutions.

Projects

RepoMind AI — Agentic RAG GitHub Repository Intelligence | *Python · LangChain*

August 2026

· ChromaDB · Sentence Transformers · Groq LLaMA · GitHub API · Agentic RAG · Advanced RAG

- Built an **Agentic RAG GitHub repository assistant** using LangChain, ChromaDB, Sentence Transformers, and Groq enabling multi-step codebase investigation through 4+ specialized tools.
- Enhanced a single-step RAG pipeline with 4 advanced retrieval techniques—query expansion, multi-query retrieval, re-ranking, and context compression—to improve relevant-context retrieval by an estimated 20–30% over the baseline system.
- Implemented agentic tool-based retrieval for repository, file, and language-specific searches, enabling 2–3× more retrieval paths for complex queries than the previous single-retrieval architecture.

AI Resume Optimizer | *Python · Groq (Llama 3.1) · Sentence Transformers · PyMuPDF · ReportLab*

June 2026

- Engineered an end-to-end **Generative AI Resume Optimizer** using FastAPI, Streamlit, Groq LLM, and Sentence Transformers for resume parsing, ATS analysis, semantic matching, and AI-powered content optimization.
- Improved job-description keyword alignment by approximately 20% through semantic similarity analysis and AI-generated resume recommendations.
- Automated resume evaluation, AI feedback generation, interview question generation, and PDF report creation, reducing manual resume review effort by over 80%.

Extracurricular Activities

Johns Hopkins Health Hackathon

- Developed a responsive frontend for an AI-powered nutrition recommendation system, enabling users to upload food images and interact with personalized dietary insights.
- Implemented frontend-to-backend API integration for submitting food images and displaying processed nutritional information and recommendation results.
- Collaborated with a multidisciplinary team to build a healthcare-focused solution combining image processing, machine learning, NLP, and Generative AI for personalized food analysis.